

PHISHING EMAIL DETECTION & AWARENESS SYSTEM

Future Interns – Cyber Security Internship

Task 2 Report

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1. Introduction

Phishing constitutes one of the most pervasive cyber threats experienced by people and companies throughout the world. Phishing involves attackers who pretend to be representatives of some trusted organization and persuade their victims to share their private data including usernames, passwords, bank logins, codes, and other personal information.

The perpetrators commonly use emails, SMS messages, or webpages designed to appear credible. Phishing predominantly focuses on human vulnerabilities since these threats have minimal effects on technical systems. Therefore, security awareness among potential victims becomes paramount.

The report discusses several examples of phishing communications, analyzes possible signs of phishing threats, defines various risks involved, and develops security awareness recommendations.

2. Objectives

The goals of this project include:

- Analyzing sample emails and messages.
- Detecting phishing signals and unusual activity.
- Determining whether messages are safe, suspicious, or phishing attacks.

- Explaining how phishing works in plain English.
 - Creating awareness about phishing for individuals and organizations.
 - Illustrating the significance of email verification and validation.
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3. Sample Analysis 1 – Legitimate Dropbox Email

Overview

The first example is a genuine Dropbox email related to account storage.

Observations

Indicator	Observation
Sender Domain	dropboxmail.com
Brand Consistency	Present
HTTPS Links	Present
Professional Language	Present
Urgency Tactics	Absent
Credential Request	Absent

Risk Classification

SAFE

Analysis

The email comes from Dropbox's official mailing system. The language used is professional, and the email doesn't solicit any passwords or personal information. The link in the email is secured using HTTPS.

Conclusion

This communication is considered legitimate and does not exhibit phishing characteristics.

4. Sample Analysis 2 – Verification Code Scam

Overview

The second example is the solicitation for the verification code.

Observations

Indicator	Observation
Verification Code Request	Present
Social Engineering	Present
Attempt to Bypass MFA	Present
User Pressure	Present

Risk Classification

PHISHING

Analysis

The attacker tries to get the two-factor authentication code from the target individual. The passwords could be safe even when verification codes are shared; this is a way through which attackers could bypass account security.

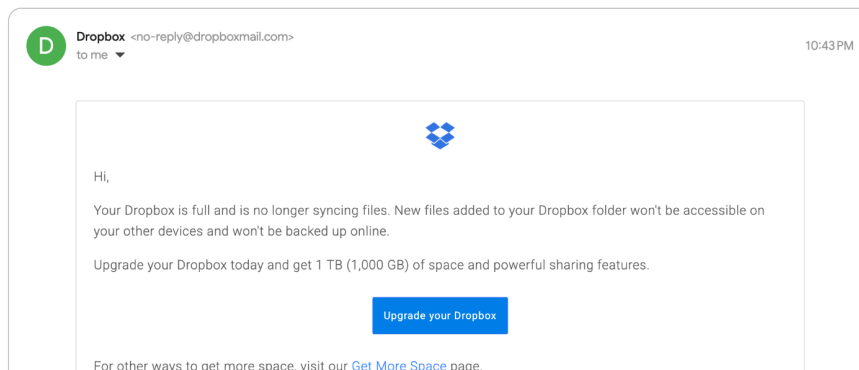
Conclusion

This message is clearly a phishing scam.

Correct! This is legitimate.

This is a legitimate Dropbox communication. The sender is "dropboxmail.com," which is unusual but legitimate, and the URL is a secure link (https) to "dropbox.com."

Show me



5. Sample Analysis 3 – Fake Google Security Alert

Overview

Sample number three masquerades as Google and asserts that somebody has stolen the password of the user.

Observations

Indicator	Observation
Look-Alike Domain	Present
Fear-Based Language	Present
Security Alert Theme	Present
Fake Sender Identity	Present

Risk Classification

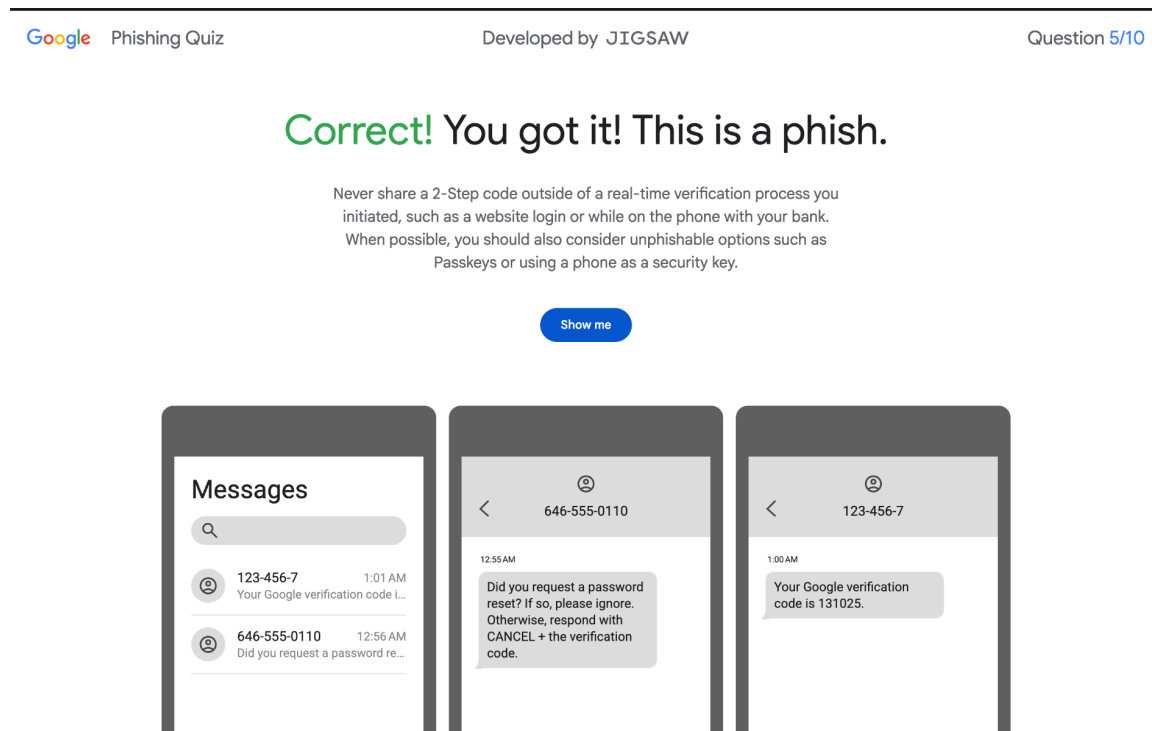
PHISHING

Analysis

It generates fear by stating that there is a threat of the user's security being compromised. Threats are often used by attackers to prompt an immediate response.

Conclusion

This message falls into the category of phishing as it involves social engineering techniques and false sender identity.



6. Sample Analysis 4 – Fake Government-Attacker Warning

Overview

The example highlights an effort by government-sponsored hackers seeking to obtain credentials of the targeted user.

Observations

Indicator	Observation
Fake Sender Domain	Present
Credential Harvesting Attempt	Present
Social Engineering	Present
Fake Login Page	Present

Risk Classification

PHISHING

Analysis

These hackers employ sophisticated social engineering techniques, creating very convincing messages to trick victims into visiting malicious webpages.

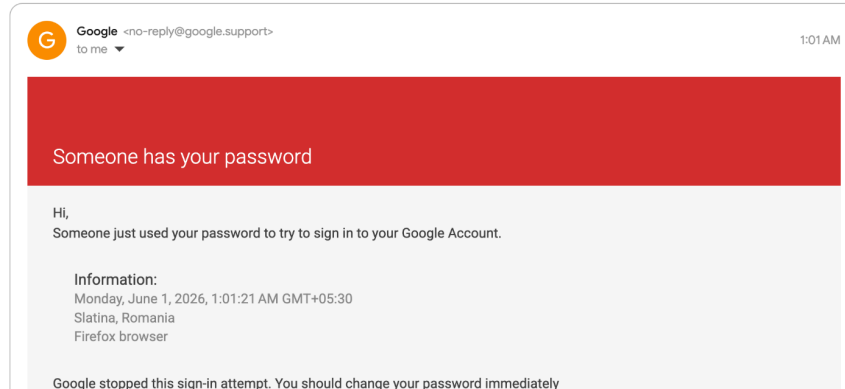
Conclusion

The message shows how effective phishing can be and must not be trusted.

Correct! This email used a look-alike URL.

This is almost identical to an attack used to successfully hack politicians' emails. Always be sure to check URLs carefully!

Show me



7. Email Header Analysis Using MXToolbox

Overview

Email headers contain key details about the source, verification, and path taken to deliver the email.

Results

Security Check	Status
SPF Alignment	Pass
SPF Authentication	Pass
DKIM Alignment	Pass
DKIM Authentication	Pass

Blacklist Status	Clean
Delivery Status	Successful

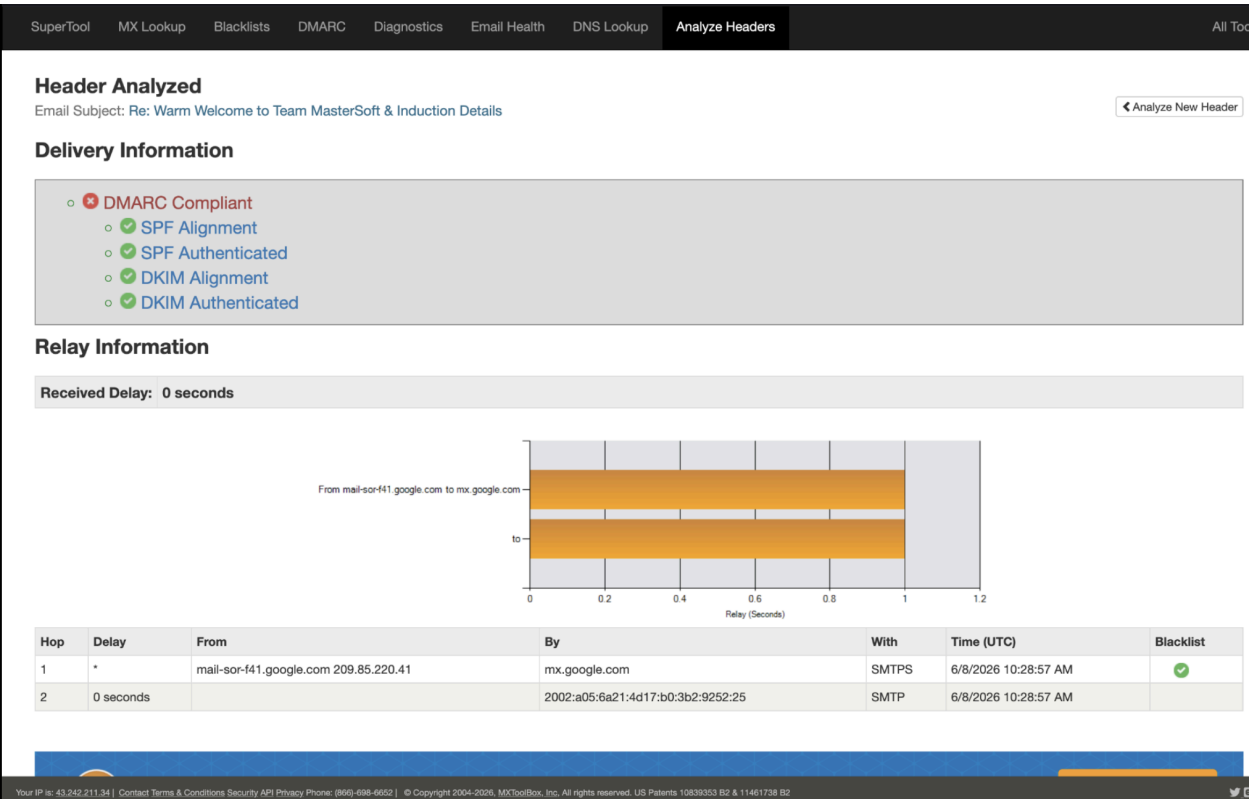
Analysis

The analyzed email is a verified email. The two processes, SPF and DKIM verification, assist in ensuring that the email comes from an authorized server without any modifications.

There are no blacklisted messages, and the email uses Google Mail as its delivery platform.

Conclusion

From the analysis of the email authentication process, it can be concluded that the email is genuine and there is no indication of spoofing.



8. Common Phishing Indicators

The following are some common characteristics associated with phishing:

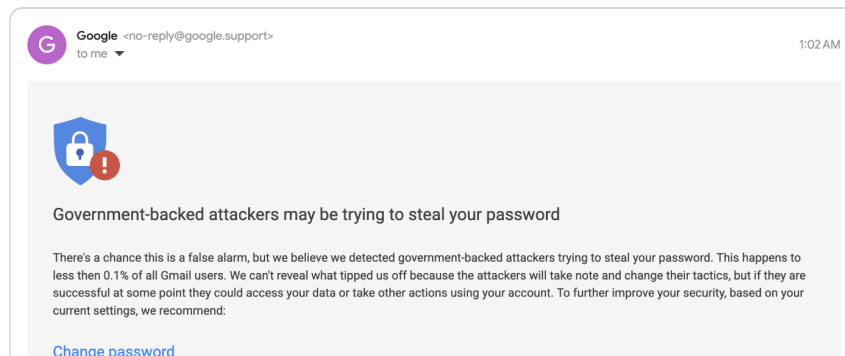
1. False sender address.
2. Domain that is similar to other sites.
3. Generic salutations like "Dear User".
4. Threatening or urgent tone.
5. Password or one-time password (OTP) requests.
6. Phishing link.
7. Strange attachment.
8. Grammar and spelling errors.
9. Requesting private information.
10. Time pressure tactics.

It is advisable for users to watch out for these signs when responding to any e-mails.

Correct! This is based on a real warning
but links to a fake login page.

The hackers tried to use Google to hide the actual link, which is from
tinyurl. An email similar to this was used to target think tanks and
politicians.

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9. Prevention & Awareness Guidelines

Do

- Verify sender email addresses.
- Hover over links before clicking.
- Use Multi-Factor Authentication (MFA).
- Keep software updated.

- Report suspicious emails immediately.
- Verify sensitive requests through official channels.

Don't

- Share passwords.
- Share OTPs or verification codes.
- Click unknown links.
- Download unexpected attachments.
- Respond to suspicious emails.
- Trust urgent requests without verification.

Google Phishing Quiz

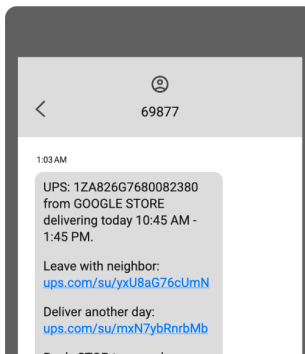
Developed by JIGSAW

Question 10/10

Correct! You got it! Go ahead and delay that package.

SMS scams are getting more common, but these links are real.

Show me



10. Employee Security Checklist

Employees need to ensure that the following questions are answered before opening an email:

- Can I trust the sender?
- Is the domain used by the sender legitimate?
- Is the email inducing an urgent response when it is not required?
- Are there requests for confidential information in the email?

- Are the links legitimate?
- Have I expected to receive such email?
- Does the email have any suspicious attachment?

If any answer raises concern, the email should be reported for further investigation.

11. Conclusion

Phishing has proven to be an effective form of attack on computers due to the way that it exploits human behavior and not technical weaknesses. The effectiveness of various strategies, including sender verification, domain validation, URL analysis, and email authentication, has been highlighted through the comparison of the two sets of emails. Security awareness is critical for mitigating phishing threats, since phishing focuses on exploiting human weaknesses rather than technical flaws. This is proved by analyzing legitimate and fraudulent samples of communication in this report to show how phishing attempts can be detected through sender authentication, domain validation, message examination, and email authentication.

Security awareness campaigns should continue in order to train users to be vigilant in verifying their messages. Awareness will remain one of the best ways to combat phishing.